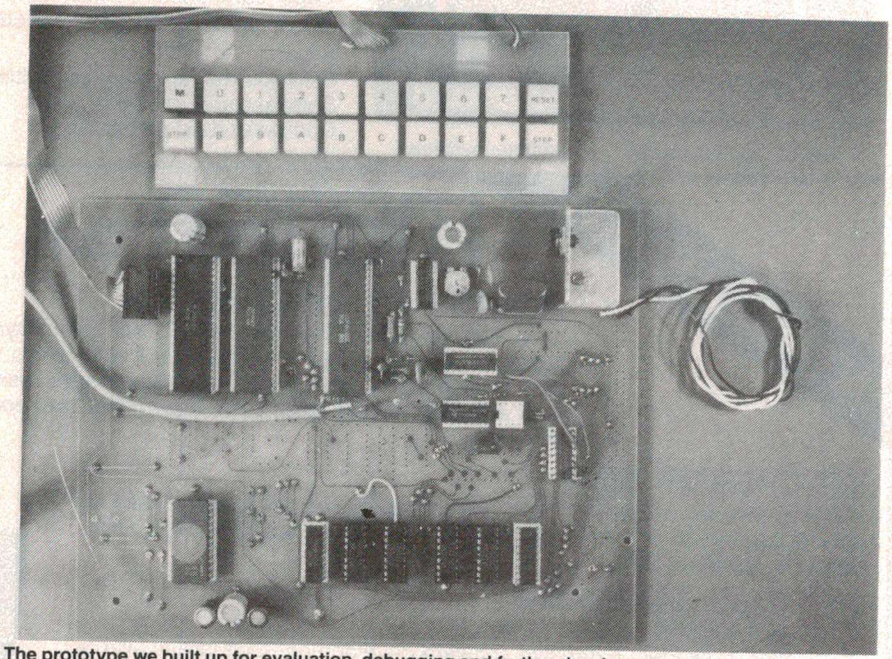
Want to get into microcomputing without boiling your brain cells or breaking the bank ?

Part 1

We often receive letters from readers that say "... I want to get into microcomputing but can't afford hundreds of dollars and don't have an engineering degree". When approached with this project, we knew that here was an answer. This project is not a 'toy' or anything like an 'evaluation kit', yet is inexpensive and simple to build and get going. It was designed by Hugh Anderson and developed for publication by Graeme Teesdale, both of whom hail from the land of the kiwi and the great white cloud.

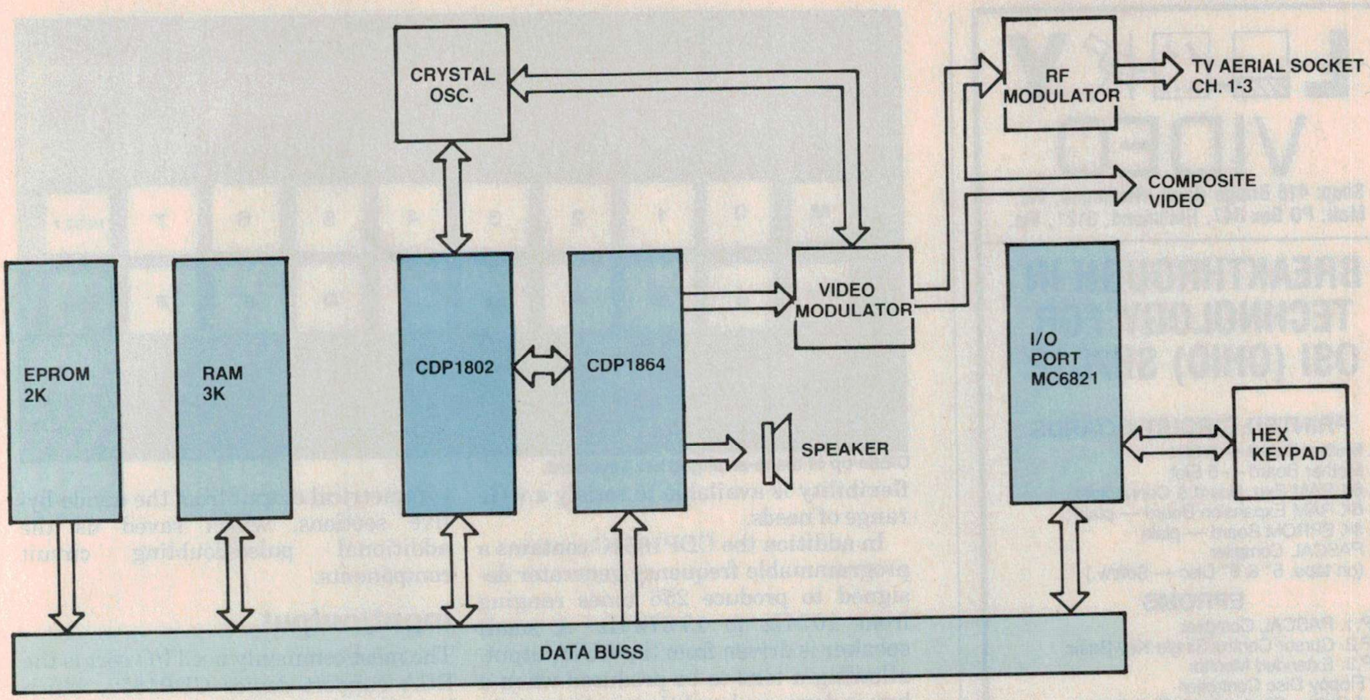
SINCE MICROPROCESSORS appeared, the concept of having a personal computer to do one's bidding has firmly taken hold in many people's minds. Electronics enthusiasts generally have a different motivation in that they're interested in the technology as well as what it will do. But the price of getting involved is generally pretty high and the level of knowledge required up front is generally pretty high too. Whilst you can buy an up-and-running computer for around $300 (the Sinclair ZX80), involvement with the hardware is minimal and many enthusiasts seek to build a computer from a kit. We've described computer projects in the past and it's possible to make up a functioning system from our ETI-680 CPU board with the addition of some peripheral hardware; the cost is around $400 but this presents a barrier to many otherwise interested enthusiasts. There are other kits around, but the price is much the same or can extend to two or three times that.

The 'evaluation kits' marketed by various microprocessor manufacturers are a less expensive alternative, but are intended for engineers and technicians and a certain level of knowledge and experience is assumed in their presentation. They also have a number of other limitations that cause many enthusiasts to shy away from them.



The prototype we built up for evaluation, debugging and further development. At top is the re-arranged hex keyboard designed to make data and program entry easy — it's also great when playing games! The final unit has a somewhat larger pc board which includes an RF modulator.

This project is aimed at people wishing to make a start in the microprocessor field, but who have no real idea where to begin. That 'dream' computer is here. The basic unit is constructed on a single pc board, but provision for expansion units — such as additional memory, standard keyboard, etc — has been included. The basic unit provides for data and program entry from a hex keypad and either a standard type or one we've especially designed for user convenience may be used. Video



can be taken direct to a monitor, or RF output from an on-board modulator can be plugged directly into the antenna of a TV set tuned to channel 1. A cassette interface is provided on board and any standard cassette recorder and tape may be employed to permanently save programs. Colour video can be obtained with the addition of further components to the basic board. An audio output, via a small speaker, is provided and is used to give an audible 'prompt' when operating the keypad. In addition, it is possible to play tunes and provide games sound effects.

Following is a general technical explanation and description of the project.

The microprocessor

The Computer is a versatile single-board mini-computer using an LSI COS/MOS CDP1802 microprocessor. This is one of two produced by RCA and is an eight-bit register-oriented central processing unit. Its main advantages are low power consumption and a wide operating voltage range of 4-10 volts.

However, one of the side effects of the low power consumption is its slow-speed machine cycle; the oscillator input frequency is 1.773 MHz to the CDP1802, machine cycle duration being approximately 4.5 μS. However, this apparently slow speed is counteracted by efficient software design.

The CDP1802 has a total of 91 easy-to-use machine code instructions, and to speed up writing programs a resident CHIP-8 language interpreter is located in EPROM.

A saving on hardware is achieved by locating the monitor and CHIP-8 interpreter in memory location 0000 to 0400. To start the monitor program running the on-board RESET button is depressed; when it is released the processor resets some of the internal registers and looks to address 0000 for the next instruction. There is thus a saving in the hardware necessary to bootstrap the processor to some other area of the memory map, e.g: 8000 to 81FF, as in some other systems. The following is a memory map for this computer:

0FFF 0C00	— additional user RAM
0BFF 0800	— additional user RAM
0700 0600	— program storage area in RAM
0500 0400	— video display memory — variables and scratchpad
0300 0200 0100 0000	— monitor and CHIP-8 interpreter located in EPROM

The user RAM is standard 2114 RAM, the final design being expanded to 3K RAM and 2K of EPROM on board. The use of the more expensive CMOS 2114 RAM will result in a lower total power consumption, and using a total of 3K standard RAM and other COS/MOS devices the total load current for the complete project is around 400 mA. This low power consumption allows the use of a simple power supply driven by an ac plug pack — no lethal 240 V mains to connect up!

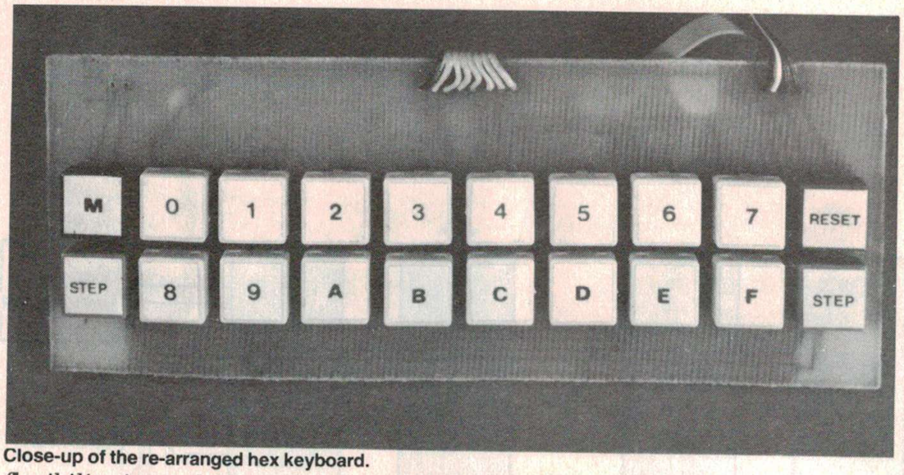
Using the cheaper 2K 2716 EPROM only a single 5 V supply is required. Only 1K of the 2716 is used at this stage, but a 'tiny BASIC' is at present under evaluation.

● This project was originally developed by Hugh Anderson for Kitparts N.Z. and known as the 'HUG 1802'. Graeme Teesdale, working in ETI's laboratory, has further refined it in conjunction with advice from Hugh, and prepared it for presentation as an ETI project, number ETI-660.

Features

RCA produce a wide range of buss-compatible devices for use with the CDP1802. For this project we have selected the CDP1864C, an LSI CMOS colour or black and white PAL-compatible video controller. The CDP1864C generates vertical sync, horizontal sync and composite sync. These signals combined with the RED, BLUE, GREEN, BURST and BACKGROUND output signals can be used to generate a composite video signal for a video monitor or into our RF modulator to your TV aerial inputs. The DMA (Direct Memory Address) feature of the 1802 is used for direct data transfer of luminance information for display refresh.

The completed toy computer with additional components added to the motherboard and under program control will produce limited background and foreground colours, but sufficient ▶



Close-up of the re-arranged hex keyboard.

flexibility is available to satisfy a wide range of needs.

In addition the CDP1864C contains a programmable frequency generator designed to produce 256 tones ranging from 107 Hz to 13 672 Hz. A small speaker is driven from the AUD output, allowing a tone to be produced when a key is depressed and the computer is in the tape load/dump routine. This generator can be used with suitable software in users' programs.

To produce the correct line and vertical sync pulse periods, a crystal frequency input of 1.75 MHz is required to the CDP1802; conversely, to produce the correct colour phase angles, a reference of twice colour-burst frequency is required by the colour modulator circuitry. Considerable attention was given to this area in an effort to reduce the crystal count to one. The frequency of the colour burst is far more important than the line or vertical frequencies, and most video or TV receivers have at least a 5% catching range about their nominal frequencies. A decision was therefore made to use the more commonly available 8.86 MHz crystal, which was divided by five to produce a frequency of 1.773 MHz. The error when compared to the original 1.75 MHz is slightly over 1%, and results in a vertical sync frequency of 50.6 Hz and a line frequency of 15 820 Hz. The only effect on screen is the 0.6 Hz beat pattern on the picture if an older monochrome receiver with a poorly filtered power supply is used. In modern colour TV receivers the effect is not noticed due to the improved power supply designs. The cost saving of not having the additional crystal and TTL oscillator outweighs any disadvantages encountered.

In practice it is difficult to obtain a symmetrical output divide-by-five to drive the clock input of the CDP1802. We experimented with different circuits until we discovered that the CDP1802 would accept an inverted non-symmetrical output from the divide-by-five sections, which saved us the additional pulse-doubling circuit components.

Input/output

The most commonly used I/O port is the RCA support device CDP1852, which has the disadvantage of not being software programmable to be input or output; the CDP1851 does not have this disadvantage but is more expensive. The more commonly available Motorola MC6821 Peripheral Interface Adaptor (PIA) was therefore chosen for the I/O port. The basic system uses only one of the two eight-bit bi-directional ports, the other being user-available, although only using machine language programming.

The PAO-PA7 section of the PIA is used to interface the hex keypad to the CPU, the other PB section not being initialised in the monitor program. The 6821 PIA is normally memory-mapped in the 6800 system, a combination of the decoded address lines and the 1802 encoded 'N' lines being used to select it for software programming.

The PIA allows a high degree of flexibility, but does require some initialisation software in the monitor for it to act as a keypad reader. Our design is arranged so that no load key is required in conjunction with the keypad to debounce keys. An audible tone is an indication of a key being depressed.

Whetted your appetite?

Construction details of this brilliant little computer will commence next issue, so clean up your soldering irons and clear a space on the workbench, kitchen table, etc. Following the construction we intend to present an article or two on programming along with a few already-developed programs that you can play with. Now you can learn about the ins and outs of microprocessing without boiling your brain cells or breaking the bank. Don't go away! ●